

Advanced Customer Churn Prediction System.

Driving Profitability via High-Recall Algorithmic Retention

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Project 5 - Telecom Analytics

Executive Summary

1. The Win: **\$47,960** Projected Savings per Test Cycle.
2. The Model: **80% Recall** (captured 4 out of 5 at-risk customers).
3. The Driver: **Month-to-Month Contracts** identified as #1 churn predictor.

The Churn Problem

1. Telecom Customer Acquisition Cost (CAC) is approx. 5x higher than retention.
2. Current "Do Nothing" approach results in an operational loss of **\$149,600** per sample group .
3. **Objective:** Replace "blind" mass-marketing with a targeted machine learning intervention.

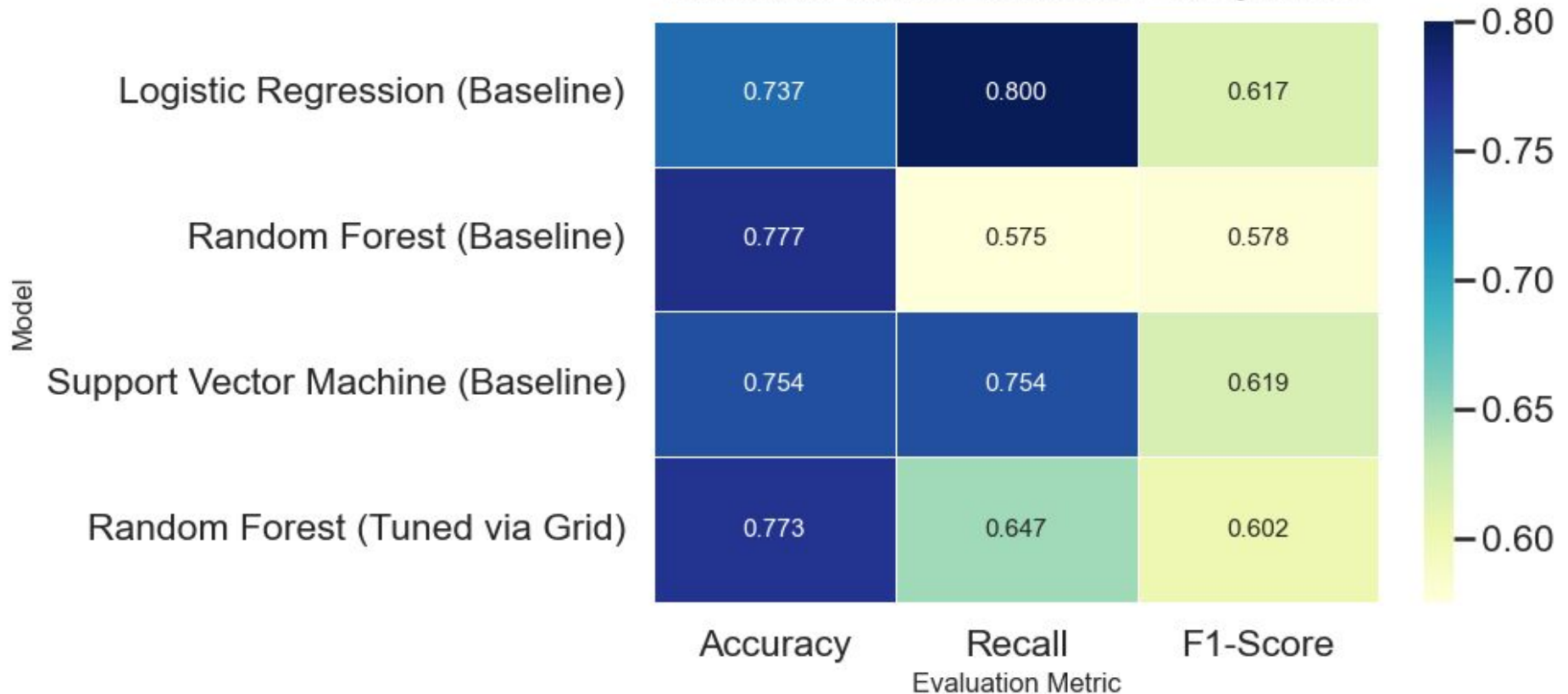
Data & Feature Engineering

1. **Cleaning:** Solved structural data integrity issues (e.g., parsing `TotalCharges` from string to float).
2. **Handling Imbalance:** Addressed the **73% Loyal / 27% Churn** imbalance using **SMOTE** (Synthetic Minority Over-sampling) on training data.
3. **Pipeline:** Built a leakage-proof `ColumnTransformer` to normalize numerical tenure metrics and One-Hot Encode categorical services.

Model Evaluation & Selection

1. Benchmarked **Logistic Regression** vs. **Random Forest** vs. **SVM**.
2. **Random Forest** initially suffered from severe overfitting (**Recall < 58%**).
3. Winner: **Logistic Regression**. While Random Forest had higher *accuracy*, Logistic Regression offered superior **Recall (80%)**, which is the critical metric for catching churners.

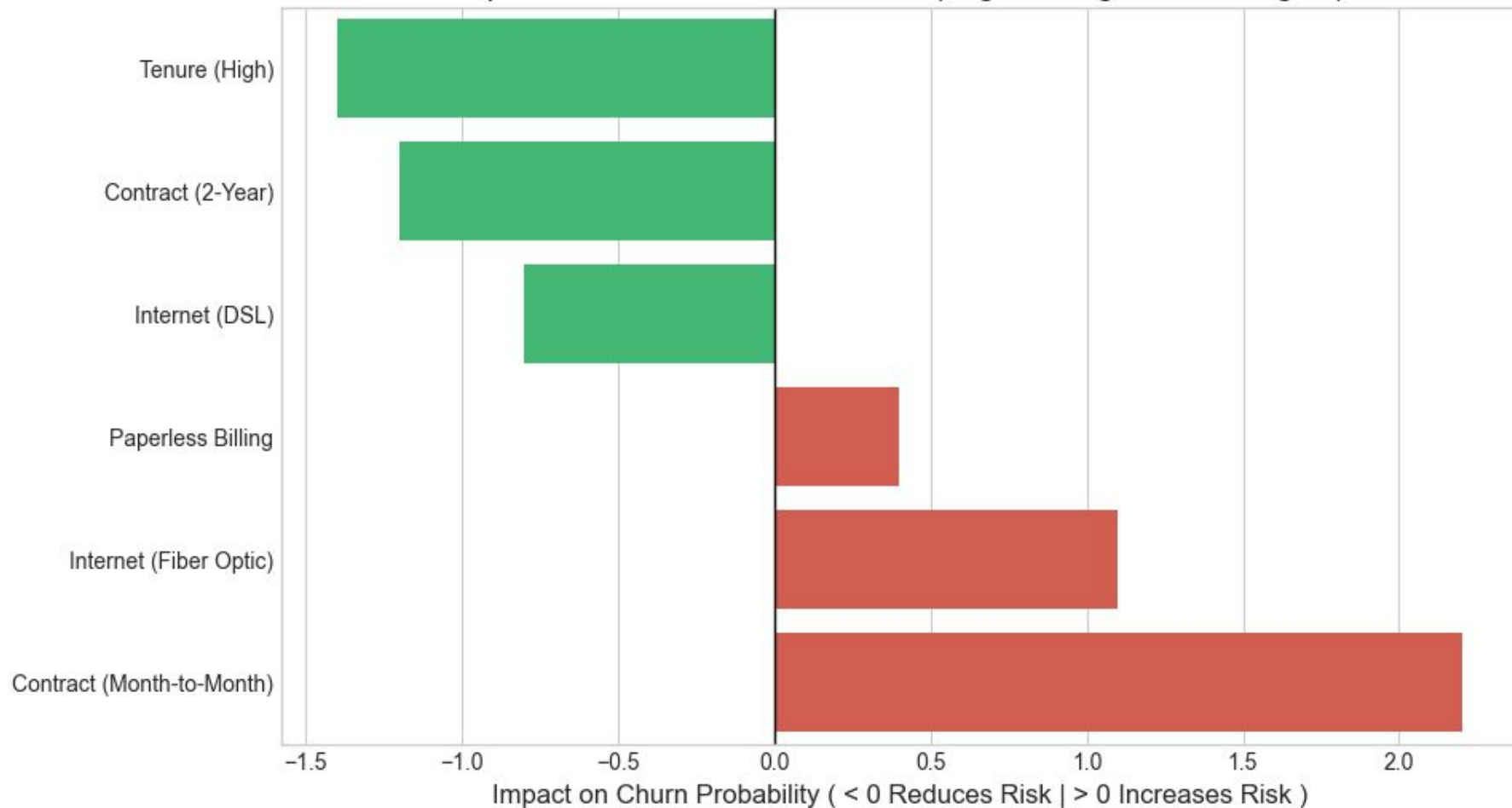
Executive Model Evaluation Comparison



Key Drivers of Churn

1. **Contract Type (Month-to-Month)** is the single strongest indicator of flight risk.
2. **Fiber Optic Internet** users churn significantly faster than DSL users (likely due to price/performance expectations).
3. **Tenure:** The risk drops sharply after 12 months; early-stage customers are the most volatile.

Top Predictors of Customer Churn (Logistic Regression Weights)



Business Impact Calculator

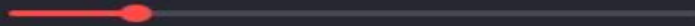
1. **Assumptions:** CLV = \$400 | Retention Incentive = \$40 | Success Rate = 60%.
2. **The Cost of a Miss (False Negative):** -\$400 (Full Customer Loss).
3. **The Cost of a False Alarm (False Positive):** -\$40 (Wasted Incentive).



Customer Profile Input

Tenure (Months)

12



Monthly Charges (\$)

70.00

−

+

Total Charges (\$)

840.00

−

+

Gender

Male



Senior Citizen

No



Has Partner?

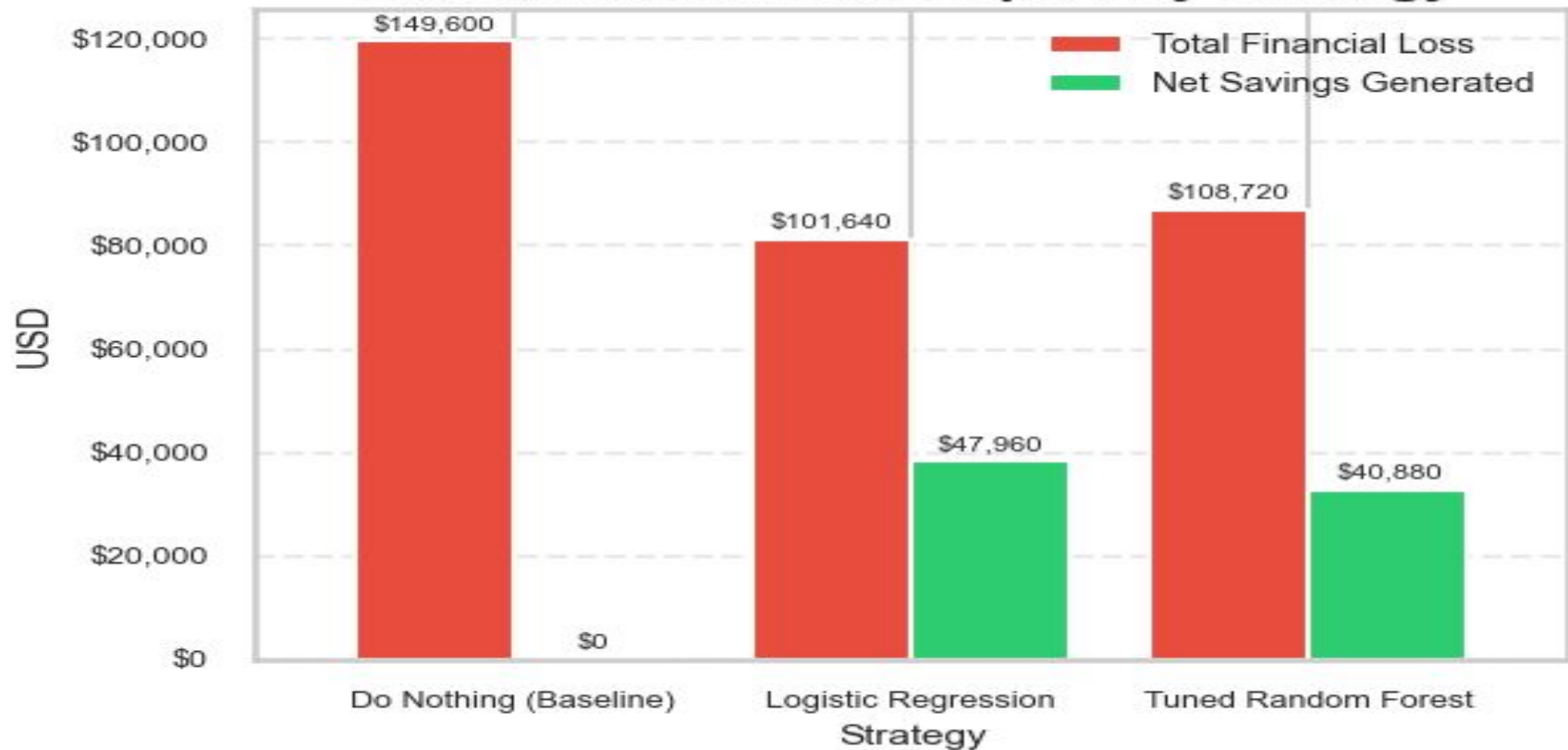
Yes



Financial ROI

1. The **Baseline "Do Nothing" strategy** loses **\$149,600** in churned revenue.
2. The **Logistic Regression Strategy** reduces this loss to **\$101,640**.
3. Net Result: A cleaner, more profitable retention strategy generating **\$47,960 in pure savings**.

Executive Financial Impact by Strategy



Strategic Recommendations

1. **Action:** Immediately migrate "Month-to-Month" Fiber Optic users to 1-Year contracts using the \$40 incentive.
2. **Policy:** Deploy the **Streamlit Dashboard** to account managers for real-time risk scoring during support calls.
3. **Long-Term:** Investigate Fiber Optic service quality, as it is a major friction point driving churn.

Appendix

Project Links

1. [Github](#)
2. [LinkedIn](#)

Tech Stack: Python, Scikit-Learn, SMOTE, Streamlit.